

## TEST 1

**Course:** Applied Statistics for Engineering and Scientists 1    **Course ID:** STAT451

**Duration:** 90 minutes

**Materials:** One A4 formula sheet allowed

**Student name:** ..... **Student ID:** ..... **No.:** .....

**Note:**

- All calculations must be rounded to **four** decimal places.
- Standard normal  $\mathcal{N}(0, 1)$  and Student  $t$  distribution tables are provided.

**Question 1.** (2.0 points) A machine produces metal sheets whose thickness (in millimeters) is normally distributed with mean  $\mu = 2.40$  mm and standard deviation  $\sigma = 0.12$  mm. Quality engineers regard a sheet as *defective* if its thickness falls outside the interval  $[2.20, 2.60]$  mm.

- Find the probability that a randomly selected sheet is defective.
- Suppose that a quality inspector checks a batch of  $n = 50$  randomly selected sheets. Using the normal approximation to the binomial distribution, compute the probability that the batch contains *at least* 4 defective sheets.
- The plant produces thousands of sheets each day. A random sample of  $n = 36$  sheets is taken, and  $\bar{X}$  denotes the sample mean thickness. Determine the probability that  $\bar{X}$  lies within 0.03 mm of the process mean.

**Question 2.** (2.0 points) The compressive strength (in MPa) of  $n = 20$  randomly selected concrete cylinders produced by a supplier yielded the following summary: sample mean  $\bar{x} = 33.48$  MPa and sample standard deviation  $s = 3.25$  MPa. Assume the population distribution is approximately normal.

- Construct a 95% confidence interval for the mean compressive strength  $\mu$ .
- A separate inspection of 400 cylinders revealed that 352 of them met a minimum strength specification. Construct a 99% confidence interval for the true proportion  $p$  of cylinders that meet this specification.
- A civil engineer wishes to estimate  $\mu$  within a margin of error of 1.0 MPa with 95% confidence. Using the preliminary estimate  $s = 3.25$  MPa, determine the minimum sample size required.

**Question 3.** (2.0 points) A manufacturer claims that the mean lifetime of its new automotive battery is at least 60 months. To test this claim, a consumer agency selected a random sample of  $n = 45$  batteries and found a sample mean lifetime of  $\bar{x} = 57.9$  months with a sample standard deviation of  $s = 6.4$  months.

- At the  $\alpha = 0.05$  significance level, do the data provide sufficient evidence to refute the manufacturer's claim? State the hypotheses, test statistic, and conclusion.

- (b) In a separate market survey, 72 out of 300 randomly selected customers reported having to replace the battery within the warranty period of 24 months. The manufacturer had advertised that *at most 20%* of batteries fail within the warranty period. Test at  $\alpha = 0.05$  whether the data contradict this claim.
- (c) Compute the  $p$ -value for part (b) and interpret it.

**Question 4.** (1.5 *points*) A new drying additive is being evaluated for latex paint. The drying times (in hours) are approximately normally distributed. The data are summarized below:

|                               | Sample size | Sample mean        | Sample std. dev. |
|-------------------------------|-------------|--------------------|------------------|
| Paint <i>without</i> additive | $n_1 = 10$  | $\bar{x}_1 = 4.85$ | $s_1 = 0.42$     |
| Paint <i>with</i> additive    | $n_2 = 12$  | $\bar{x}_2 = 4.22$ | $s_2 = 0.51$     |

Assume both population variances are equal.

- (a) Test at  $\alpha = 0.05$  whether the additive significantly reduces the mean drying time by more than 0.3 hours.
- (b) Construct a 95% confidence interval for  $\mu_1 - \mu_2$ , the true difference in mean drying times.

**Question 5.** (2.5 *points*) A materials engineer studies the relationship between the curing temperature  $x$  (in °C) and the tensile strength  $y$  (in  $10^3$  psi) of a polymer. The data for  $n = 10$  specimens are:

|     |      |      |      |      |      |      |      |      |      |      |
|-----|------|------|------|------|------|------|------|------|------|------|
| $x$ | 90   | 100  | 110  | 120  | 130  | 140  | 150  | 160  | 170  | 180  |
| $y$ | 45.1 | 47.8 | 49.3 | 51.7 | 52.9 | 55.0 | 56.2 | 58.1 | 59.4 | 61.3 |

- (a) Determine the estimated least-squares regression line  $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$  and interpret the slope in context.
- (b) Compute the sample correlation coefficient  $r$  and interpret it.
- (c) Compute the coefficient of determination  $R^2$  and interpret its meaning.
- (d) Predict the tensile strength when the curing temperature is  $x = 155^\circ\text{C}$  and when  $x = 250^\circ\text{C}$ . For each prediction, comment on whether extrapolation is a concern.

— END OF TEST 1 —